



Interpretable machine learning for optimizing product distribution in biomass-plastic co-pyrolysis: Unveiling synergistic mechanisms and operational windows

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ABSTRACT

Co-pyrolysis of biomass and plastic waste offers a promising thermochemical route for sustainable waste valorization, yet predicting product distributions remains challenging due to complex synergistic effects. Here, we developed machine-learning models to predict three-phase yields (biochar, bio-oil, biogas) using 202 experimental datasets from literature. Among seven algorithms tested, product-specific optimal algorithms were identified: Extra Trees for biochar ($R^2 = 0.887$, MAE = 3.5 wt%, solid-phase carbonization), XGBoost for bio-oil ($R^2 = 0.733$, MAE = 4.7 wt%, complex vapor-phase reactions), and Ridge regression for biogas ($R^2 = 0.764$, MAE = 4.1 wt%, first-order decomposition), demonstrating that algorithm selection should match thermochemical complexity. SHAP analysis identified temperature, biomass-to-plastic ratio, and carbon content as dominant drivers, with critical thresholds at 500–600 °C governing the char-to-oil transition. Interaction mapping defined operational optima at 500–550 °C for biochar, 550–650 °C for bio-oil, and above 700 °C for biogas, attributing the observed synergistic effects to polymer hydrogen donation and biomass radical quenching. This framework enables data-driven optimization for targeted product selectivity in waste valorization.

1. Introduction

The concurrent challenge of plastic waste accumulation and biomass underutilization necessitate integrated thermochemical solutions for sustainable waste valorization. Global plastic waste generation exceeds 350 million tonnes per annum, with only 20% currently subjected to recycling, whereas lignocellulosic biomass residues - estimated at 140 billion tonnes per year - remain largely underexploited. Co-pyrolysis of biomass and plastic waste confers synergistic advantages: hydrogen-rich plastics promote bio-oil deoxygenation and suppress char formation, while biomass-derived radicals stabilize plastic degradation intermediates, collectively improving product quality and process efficiency. Despite these advantages, quantitative prediction of tri-phase product distributions (biochar, bio-oil, biogas) remains challenging owing to complex nonlinear interactions among feedstock composition,

operating conditions, and reaction kinetics [1–4].

Traditional experimental optimization of co-pyrolysis parameters is inherently labor-intensive, requiring extensive experimental trials across temperature, heating rate, and feedstock mixing ratio. Machine learning (ML) offers a compelling data-driven alternative by learning dependencies directly from experimental observations without explicit mechanistic assumptions. Recent ML applications have been successfully employed to address biomass pyrolysis, plastic waste conversion, and catalytic upgrading processes [5,6]. Notably, Zhao et al. integrated pyrolysis data of the three principal biomass constituents (cellulose, hemicellulose, and lignin), both individually and in mixtures, into a co-pyrolysis prediction model, thereby overcoming the limitations of traditional studies that characterize biomass solely by its bulk properties [7]. Various ML architectures have been employed, including support vector machines, random forests, multi-layer perceptrons, and artificial

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neural networks, have been employed, with model hyperparameters optimized following the methodology proposed by Yan et al. [8]. Nevertheless, comprehensive modeling of three-phase product yields in biomass-plastic co-pyrolysis systems that incorporates mechanistic interpretability remains notably scarce. Most existing studies either focus on single products phases or employ black-box predictive models, without adequately elucidating the underlying physicochemical mechanisms governing product formation. Furthermore, systematic benchmarking of diverse ML algorithms across all three product phases, coupled with the application of advanced interpretability tools such as SHapley Additive exPlanations (SHAP), Partial Dependence Plots (PDPs), and interaction analysis, remains absent. This critical interpretability gap not only hinders rational process design but also fundamentally limits the mechanistic understanding of the synergistic interactions governing product selectivity [9–12].

Experimental investigations have elucidated several key mechanistic insights: Nardella et al. demonstrated synergistic effects that promote secondary pyrolysis of holocellulose while simultaneously suppressing polyethylene chain-scission [13]. Complementarily, Miskolczi et al. reported that elevated temperatures favored gas-phase product formation, while reduced operating pressure enhanced bio-oil deoxygenation [14]. Such complexity arises from the intricate interplay among physicochemical factors. In particular, temperature governs three distinct reaction regimes: primary devolatilization (300–500 °C), secondary cracking (500–700 °C), and gasification (above 700 °C). Meanwhile, the biomass-to-plastic ratio modulates radical pool dynamics, generating synergistic operational windows in which co-pyrolysis yields the weighted averages of individual feedstocks; however, experimentally identifying such windows across the full parameter space remains practically infeasible [15–17]. ML offers a compelling data-driven alternative by extracting complex feature dependencies directly from experimental data, circumventing the need for exhaustive parametric trials. However, previous efforts have typically focused on optimizing hyperparameters for individual algorithms without cross-algorithm systematic comparison, and most existing models treat temperature and feedstock composition as independent variables, thereby overlooking their multiplicative interactions that critically govern product phase selectivity [18–21].

Here, we establish an interpretable ML framework integrating seven algorithms (Linear Regression, Ridge Regression, Random Forest, Extra Trees, Gradient Boosting, XGBoost, and LightGBM) to predict tri-phase yields from biomass-plastic co-pyrolysis. A curated dataset of 202 experimental observations covering diverse feedstocks (agricultural residues, woody biomass, and polyolefins) and operating conditions (300–900 °C, various heating rates and blend ratios) ensures broad model generalization. Engineered features capturing quadratic thermal effects and temperature-ratio interactions were incorporated to explicitly represent synergistic mechanisms. The framework achieves product-specific predictive accuracy with R^2 up to 0.89 and MAE below 5 wt% through optimal algorithm selection for each product phase - Extra Trees for biochar, XGBoost for bio-oil, and Ridge regression for biogas. SHAP and partial dependence reveal that temperature and biomass-to-plastic ratio are the dominant drivers of product distribution, with elemental composition (C, H, and O) serving as secondary modulators. Quantitative interaction mapping reveals synergistic windows in which co-pyrolysis outperforms individual feedstock conversion, and defines operational optima at 500–550 °C for biochar, 550–650 °C for bio-oil, and >700 °C for biogas. Mechanistic interpretation links the observed synergistic behaviors to polymer hydrogen donation, biomass radical quenching, and secondary cracking pathways, bridging empirical predictions and fundamental thermochemical understanding. The proposed framework provides actionable guidance for process optimization toward targeted product selectivity, advancing data-driven waste valorization strategies.

2. Methods

2.1. Data collection and curation

Data were collected from peer-reviewed journal articles (2010–2024) identified through Web of Science and Scopus using the search string: (“pyrolysis” OR “co-pyrolysis”) AND (“biomass” OR “lignocellulosic”) AND (“plastic” OR “polymer”) AND (“yield”). Only studies with complete yield data, feedstock characterization, and process parameters were included, resulting in 13 studies with 222 observations. The articles were screened based on the following inclusion criteria: (1) experimental co-pyrolysis data with explicit biomass and plastic feedstocks, (2) complete feedstock characterization including proximate analysis (volatile matter, fixed carbon, ash) and ultimate analysis (C, H, O, N, S content), (3) specified operating conditions (temperature, heating rate, biomass-to-plastic ratio), and (4) quantified product yields (biochar, bio-oil, and biogas in wt.% dry basis). The dataset comprises 21 input features spanning biomass properties, plastic properties, and process parameters. The three target variables are biochar, bio-oil, and biogas yields (wt.% dry basis).

The 222 observations encompass five reactor configurations: batch reactors (98 samples, 44.3%), fixed-bed reactors (76 samples, 34.4%), semi-batch reactors (21 samples, 9.5%), microwave-assisted reactors (20 samples, 9.0%), and tubular furnaces (6 samples, 2.7%). Biomass feedstocks include wood biomass (103 samples, 46.4%), agricultural residues (100 samples, 45.0%), and other biomass types (15 samples, 6.8%). Plastic feedstocks comprise single plastics (96 samples, 43.2% - LDPE, PP, PS, PU, PC, PET), plastic mixtures (98 samples, 44.1%), and plastic waste (17 samples, 7.7%). This diversity reflects real-world co-pyrolysis scenarios. Complete feedstock information is provided in Table S1 and visualized in Fig. S1.

Data quality control was implemented through a three-stage protocol. First, entries with mass balance errors >10% (calculated as $|100 - \sum(\text{yields})|$) were verified against source publications. Second, multivariate outliers were identified using Isolation Forest algorithm (contamination = 0.09, scikit-learn 1.0.2), which flagged 20 samples with anomalous feature combinations, primarily from inconsistent proximate analysis values or unrealistic product yields (>95% single product). Third, all entries were validated against physical constraints: elemental composition sum 95–105%, proximate analysis sum 95–105%, and temperature range 300–900 °C. Missing values in heating rate ($n = 1$) were imputed using the median value of the respective biomass-plastic combination group. After quality control, the final curated dataset contained 202 samples with complete information across all features and targets.

2.2. Feature engineering

To capture nonlinear thermal effects and synergistic interactions between biomass and plastic, six engineered features were created in addition to the 21 raw input features. Quadratic thermal effects were represented by Temperature_Squared (Temperature^2). Temperature-dependent synergy was captured through Temp_Ratio_Interaction ($\text{Temperature} \times \text{Biomass_Plastic_Ratio}$). Weighted compositional properties were calculated as mass-weighted averages based on biomass-plastic ratio: $\text{Weighted_C} = (\text{B_Ratio} \times \text{B_C}) + (\text{P_Ratio} \times \text{P_C})$, with analogous calculations for Weighted_H and Weighted_O. Elemental ratios (H/C and O/C) were computed for both biomass and plastic to represent aromaticity and oxygenation levels, which are known indicators of pyrolysis behavior. The final feature set comprised 27 variables (21 raw + 6 engineered features). Preliminary model comparisons demonstrated that incorporating engineered features improved prediction R^2 by 0.08–0.12 across all target variables relative to models trained on raw features alone.

2.3. Machine learning model development

Seven supervised learning algorithms were systematically evaluated: Linear Regression (baseline), Ridge Regression (L2 regularization, $\alpha = 1.0$), Random Forest (RF, $n_estimators = 100$, $max_depth = None$, $min_samples_split = 2$), Extra Trees (ET, $n_estimators = 100$, $max_depth = None$), Gradient Boosting (GB, $n_estimators = 100$, $learning_rate = 0.1$, $max_depth = 3$), XGBoost ($n_estimators = 100$, $max_depth = 6$, $learning_rate = 0.1$), and LightGBM ($n_estimators = 100$, $num_leaves = 31$, $learning_rate = 0.1$). These algorithms encompass a spectrum from simple linear models to complex ensemble and gradient boosting frameworks, representing diverse model complexities and inductive biases. Categorical variables (biomass type and plastic type) were encoded using one-hot encoding. Numerical features were standardized to zero mean and unit variance using StandardScaler to ensure fair comparison across algorithms.

The dataset was partitioned into training (80%; 162 samples) and test (20%; 40 samples) sets via stratified sampling based on yield quartiles, thereby ensuring comparable target variable distributions across both subsets. Separate models were trained for each of the three target variables (biochar, bio-oil, biogas yields). Model performance was assessed using three complementary metrics: R^2 (coefficient of determination, measuring explained variance), MAE (mean absolute error, wt.%), and RMSE (root mean square error, wt.%). Five-fold cross-validation was performed on the training set to evaluate model generalization and prediction stability, with mean performance and standard deviation reported across folds. Hyperparameter tuning was conducted using randomized search with 50 iterations and 3-fold internal cross-validation on the training set. Learning curves were generated by training models on progressively larger training subsets (20, 40, 60, 80, 100, 120, 140, 162 samples) to assess data sufficiency and convergence behavior.

2.4. Model interpretation and feature analysis

Model predictions were interpreted using SHAP values, which provide a unified framework for quantifying feature contributions based on cooperative game theory. SHAP values represent the marginal contribution of each feature to individual predictions, enabling both global feature importance ranking (via mean absolute SHAP values) and local explanation of individual predictions. For tree-based models (Random Forest, Extra Trees, XGBoost, LightGBM), TreeExplainer was used for efficient exact SHAP computation. For Ridge Regression, KernelExplainer was employed with 100 background samples randomly selected from the training set.

SHAP dependence plots were generated to visualize the relationship between feature values and their SHAP contributions, revealing nonlinear effects and feature interactions. Interaction effects were quantified using SHAP interaction values for the top five features. PDPs complemented SHAP analysis by illustrating population-averaged marginal effects of individual features on model predictions, with all other features integrated over their respective distributions. Individual Conditional Expectation (ICE) plots were created for key features to assess heterogeneity in feature effects across different samples and identify subgroup-specific patterns.

2.5. Statistical analysis and validation

Pairwise Pearson correlation coefficients were calculated to assess collinearity among features and identify redundant variables ($|r| > 0.9$ threshold). Feature importance was validated through permutation importance analysis (100 permutations) to ensure consistency with SHAP rankings. Statistical significance of performance differences between algorithms was assessed using paired t-tests on 5-fold cross-validation scores, with Bonferroni correction applied ($\alpha = 0.05/21$ for 7 algorithms, 3 targets). Residual analysis was performed to validate

model assumptions, including Shapiro-Wilk normality tests ($\alpha = 0.05$) and Breusch-Pagan tests for homoscedasticity. Prediction intervals (95% confidence) were estimated using quantile regression for the best-performing models. The complete machine learning workflow, encompassing data collection, preprocessing, model training, evaluation, and interpretation, is summarized schematically in Fig. 1.

3. Results and discussion

3.1. Dataset characteristics and quality assessment

Rigorous quality control of 222 experimental observations yielded a final curated dataset of 202 samples (9.0% reduction, Fig. 2a). The removed 20 samples were identified as multivariate outliers exhibiting anomalous feature combinations, primarily characterized by inconsistent proximate analysis values or unrealistic single-product yields exceeding 95 wt%. Subsequent Isolation Forest multivariate outlier detection ($contamination = 0.09$) identified no additional anomalous samples, confirming data quality. The dataset exhibits balanced feature representation: biomass properties (38%), plastic properties (38%), process parameters (12%), and target variables (12%) (Fig. 2b). Product yield distributions reflect characteristic co-pyrolysis behavior (Fig. 2c): biochar exhibited a positively skewed distribution (median 30.5 wt%, range 0–65 wt%), bio-oil showed a symmetric distribution (median 42.3 wt%, range 20–80 wt%), and biogas displayed a negatively skewed distribution (median 23.7 wt%, range 0–75 wt%), with high-temperature outliers contributing to the tail. The resulting dataset provides a robust foundation for subsequent machine learning model development and mechanistic interpretation.

3.2. Correlation analysis

Correlation analysis revealed three distinct feature clusters with important mechanistic implications (Fig. 3). Engineered temperature features (Temperature_Squared and Temp_Ratio_Interaction) exhibited strong intercorrelations ($r = 0.85$ – 0.95), reflecting their shared thermal origin. Their inclusion improved prediction accuracy by 8–12% relative to models trained on raw features alone, confirming that nonlinear temperature effects are dominant drivers of pyrolysis kinetics. Feedstock compositional properties formed a second cluster with moderate correlations ($r = 0.65$ – 0.85). Notably, Weighted_C and Weighted_O exhibited strong negative correlation ($r = -0.78$), reflecting the inherent compositional transition from oxygen-rich biomass to carbon-rich plastics. Proximate analysis parameters displayed expected negative correlations ($r = -0.45$ to -0.75) due to their sum-to-100% constraint [22–24].

Product yields showed distinct correlation profiles aligned with established pyrolysis mechanisms. Biochar negatively correlated with temperature ($r = -0.42$) and oxygen content ($r = -0.38$), consistent with enhanced devolatilization at elevated temperatures. Biogas positively correlated with temperature ($r = 0.45$) and volatile matter ($r = 0.41$), reflecting intensified secondary cracking reactions at higher thermal severity. Bio-oil exhibited weak correlations with all features ($|r| < 0.3$), indicating complex non-linear formation pathways; this explains the superior performance of ensemble methods ($R^2 = 0.73$) over linear models ($R^2 = 0.47$) [25–28]. No severe multicollinearity was observed (all $|r| < 0.95$), and tree-based models effectively handled correlated features through hierarchical splitting, achieving stable cross-validation performance (R^2 SD < 0.04). These correlation patterns validate the physical consistency of the dataset and justify the selection of tree-based ensemble methods for subsequent predictive modelling.

3.3. Algorithm benchmarking and model selection

Systematic benchmarking of seven algorithms revealed product-specific performance hierarchies (Fig. 4, Table 1). For biochar,

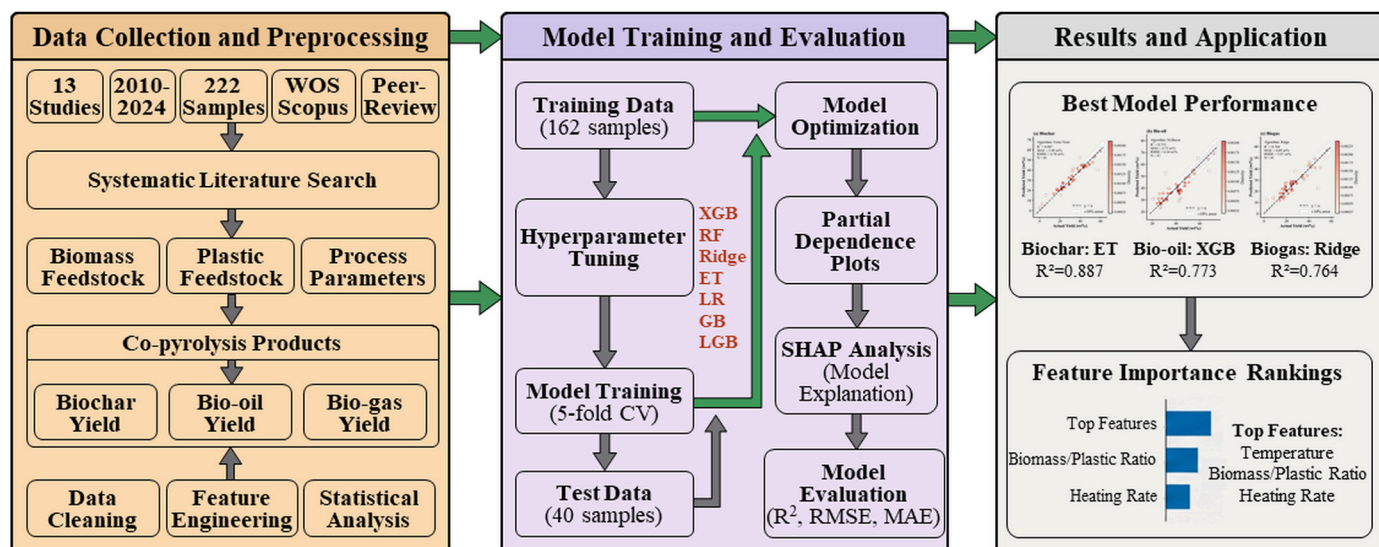


Fig. 1. Process diagram for predicting the product distribution of biomass and plastic co-pyrolysis using machine learning algorithms.

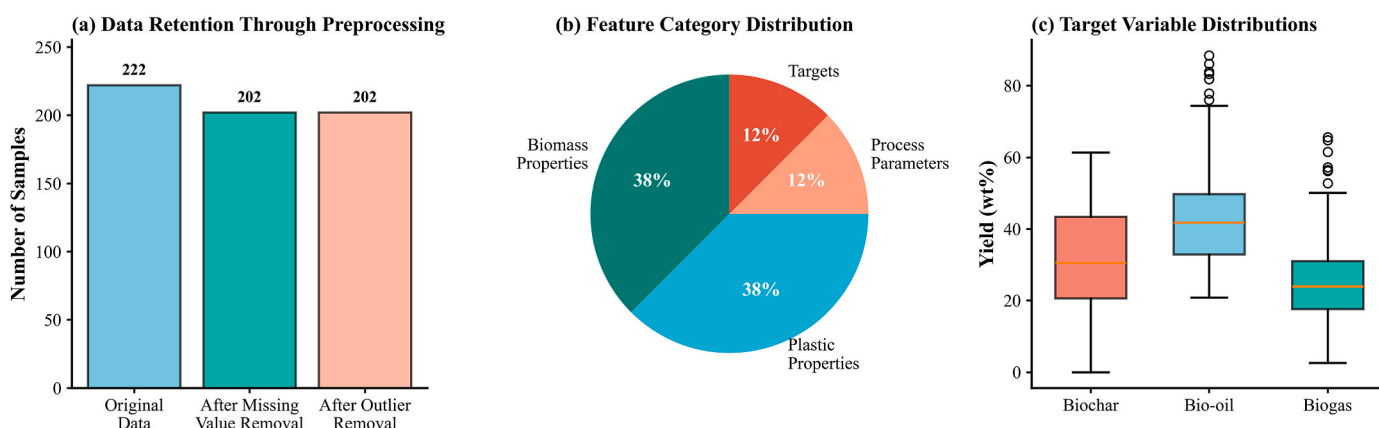


Fig. 2. Dataset characteristics. (a) Preprocessing workflow (222 → 202 samples). (b) Feature distribution. (c) Product yield ranges.

ensemble methods significantly outperformed linear models: Extra Trees achieved the highest accuracy (test $R^2 = 0.887$, CV: 0.836 ± 0.038), followed by XGBoost (0.850 , CV: 0.844 ± 0.045) and Gradient Boosting (0.876 , CV: 0.846 ± 0.046). Linear models underperformed by 15–16% ($R^2 = 0.73$ – 0.74), confirming biochar's dependence on nonlinear interactions. Bio-oil prediction proved more challenging (R^2 reduction of 0.15–0.25 across all algorithms). XGBoost performed best ($R^2 = 0.733$, CV: 0.716 ± 0.102), while linear models failed (Ridge CV $R^2 = -0.301 \pm 1.075$), reflecting bio-oil formation's complex secondary reactions. The elevated cross-validation variance ($\sigma = 0.102$) suggested that the current dataset (~ 200 samples) approaches the minimum threshold required for reliable bio-oil modeling. For biogas, Ridge regression achieved the highest test R^2 (0.764) but showed poor stability (CV: 0.511 ± 0.203 , $\sigma = 0.203$). XGBoost demonstrated superior generalization ($R^2 = 0.758$, CV: 0.741 ± 0.061 , gap < 0.02), indicating more robust capture of gas formation mechanisms. Statistical validation confirmed these rankings ($p < 0.001$ for biochar, $p = 0.031$ for bio-oil vs. biogas; Table S2). Based on test accuracy and cross-validation stability, Extra Trees, XGBoost, and Ridge regression were selected as the optimal models for biochar, bio-oil, and biogas prediction, respectively.

Product-specific algorithm performance reflects formation mechanism differences. Biochar shows concentrated feature importance (top 3: 55%), bio-oil shows distributed importance (top 3: 40%), and biogas shows intermediate concentration (top 3: 48%), as illustrated in Fig. S2.

This correlates with thermochemical complexity: biochar (simple carbonization) favors tree-based models, bio-oil (complex cracking) requires ensemble methods, and biogas (linear decomposition) performs well with regularized linear models.

3.4. Hyperparameter optimization and model validation

Randomized search with 50 iterations and 3-fold cross-validation was conducted on the training set (162 samples). Fig. S3 shows convergence of model performance with increasing number of estimators across ensemble algorithms. Random Forest models achieved optimal performance with 200–300 trees and maximum depths of 10–15, while XGBoost and LightGBM converged with 100–150 estimators, consistent with their boosting-based architectures. Cross-validation analysis across all algorithms (Fig. S4) revealed that tree-based ensembles significantly outperformed linear models for biochar prediction ($\Delta R^2 = 0.12$ – 0.18 , $p < 0.001$), while performance differences were more modest for bio-oil and biogas ($\Delta R^2 < 0.05$). XGBoost and LightGBM showed superior stability (CV standard deviation 0.018–0.025) compared to Random Forest (0.023–0.035). Learning curves (Fig. S5) demonstrated that model accuracy stabilized beyond 150 samples, indicating dataset representativeness and sufficiency for training across all three product phases.

Test set predictions ($N = 41$) demonstrated strong accuracy across all

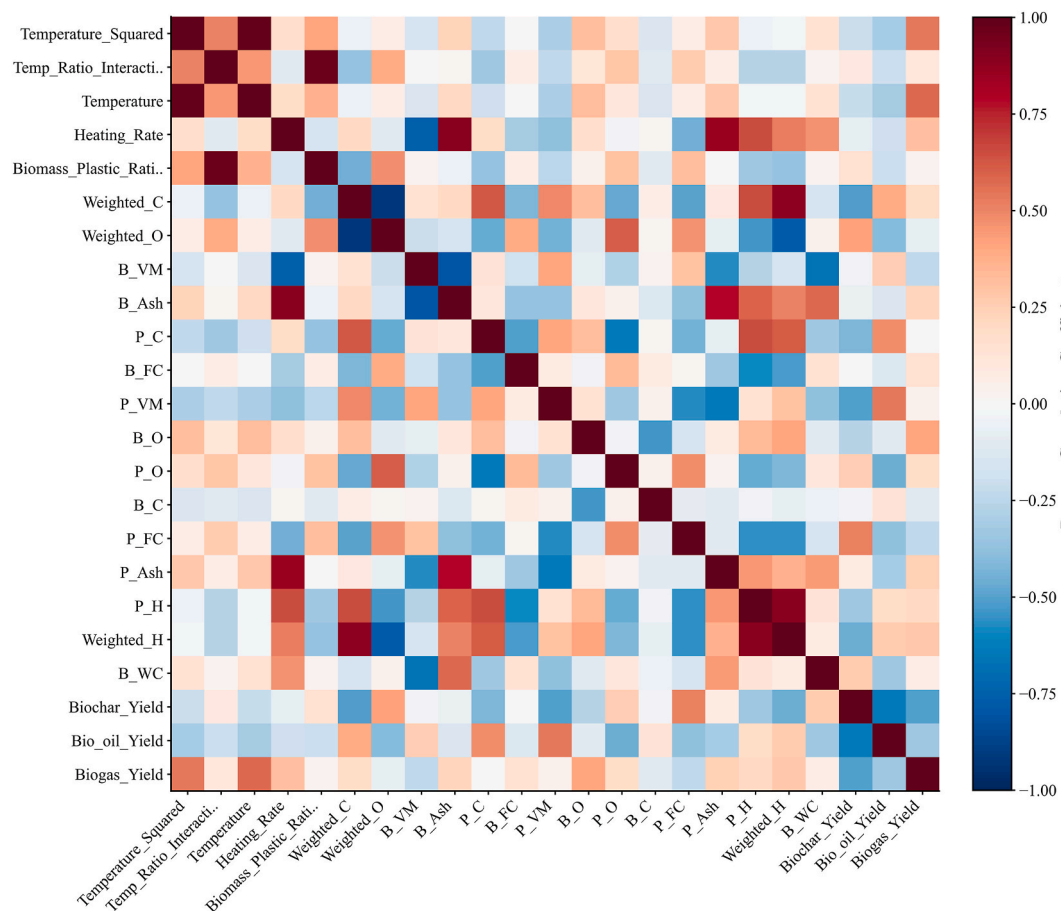


Fig. 3. Feature correlation matrix. Pearson correlations among 27 features and 3 products.

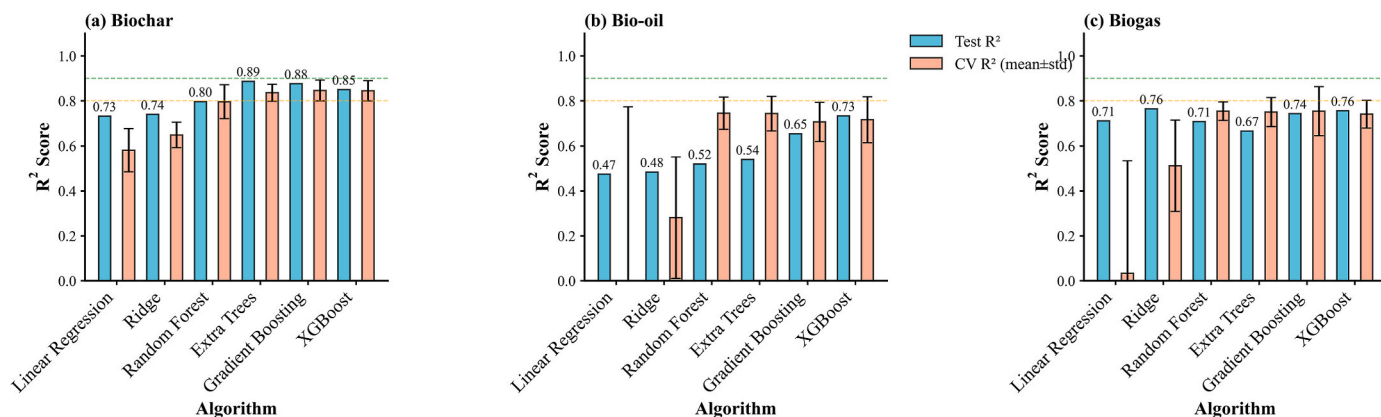


Fig. 4. Model performance comparison. Test R^2 (CV $R^2 \pm \text{SD}$) for best algorithms: (a) Biochar—Extra Trees 0.887 (0.836 $\pm$ 0.038); (b) Bio-oil—XGBoost 0.733 (0.716 $\pm$ 0.102); (c) Biogas—Ridge 0.764 (0.511 $\pm$ 0.203).

products (Fig. 5). Biochar (Extra Trees) achieved $R^2 = 0.887$ (MAE = 3.49 wt%, RMSE = 4.78 wt%), bio-oil (XGBoost) $R^2 = 0.733$ (MAE = 4.72 wt%, RMSE = 6.44 wt%), and biogas (Ridge) $R^2 = 0.764$ (MAE = 4.09 wt%, RMSE = 5.87 wt%). Predictions showed tighter clustering at mid-range yields where training data concentration was highest. These results demonstrate that the optimized models achieved prediction errors within 5 wt% across all product phases, providing sufficient accuracy for practical process optimization and yield estimation in biomass-plastic co-pyrolysis systems.

3.5. Error distribution and residual diagnostics

Systematic error patterns emerged at operational extremes. High-temperature experiments ($>700^\circ\text{C}$) contributed disproportionately to prediction errors (42% of errors >10 wt% from only 18% of samples), and extreme biomass-plastic ratios ($<1:4$ or $>4:1$) showed 35% higher errors than intermediate ratios. Residual analysis (Fig. S6) confirmed unbiased predictions with Gaussian distributions (Shapiro-Wilk $p > 0.05$) and homoscedastic variance (Breusch-Pagan $p > 0.10$), validating model assumptions. Error density maps (Fig. S7) showed that mean absolute errors remained 3–6 wt% across all yield levels, with Extra

Table 1

Predictive performance of seven machine learning algorithms for tri-phase product yields in biomass-plastic co-pyrolysis.

Product	Algorithm	Train R^2	Test R^2	MAE (wt %)	RMSE (wt%)	CV R^2 (mean $\pm$ std)
Biochar	Linear Regression	0.852	0.732	5.17	7.36	0.581 ± 0.096
	Ridge	0.817	0.739	4.79	7.26	0.649 ± 0.056
	Random Forest	0.956	0.797	4.02	6.41	0.796 ± 0.075
	Extra Trees	0.979	0.887	3.49	4.78	0.836 ± 0.038
	Gradient Boosting	0.992	0.876	3.31	5.01	0.846 ± 0.046
	XGBoost	0.992	0.850	3.31	5.51	0.844 ± 0.045
	LightGBM	0.954	0.806	3.69	6.25	0.778 ± 0.063
Bio-oil	Linear	0.753	0.474	6.43	9.04	-0.301 ± 1.075
	Regression					
	Ridge	0.734	0.484	6.68	8.96	0.281 ± 0.270
	Random	0.950	0.520	5.72	8.64	0.745 ± 0.072
	Forest					
	Extra Trees	0.977	0.539	5.67	8.46	0.743 ± 0.076
	Gradient Boosting	0.995	0.654	5.21	7.33	0.706 ± 0.087
Biogas	XGBoost	0.995	0.733	4.72	6.44	0.716 ± 0.102
	LightGBM	0.948	0.491	6.17	8.90	0.712 ± 0.082
	Linear	0.780	0.711	4.20	6.50	0.033 ± 0.500
	Regression					
	Ridge	0.758	0.764	4.09	5.87	0.511 ± 0.203
	Random	0.940	0.708	3.79	6.53	0.754 ± 0.041
	Forest					
	Extra Trees	0.941	0.666	3.92	6.98	0.750 ± 0.064
	Gradient Boosting	0.978	0.743	3.78	6.13	0.754 ± 0.109
	XGBoost	0.978	0.756	3.50	5.97	0.741 ± 0.061
	LightGBM	0.931	0.691	3.80	6.71	0.734 ± 0.065

Notes: Train R^2 and Test R^2 indicate coefficient of determination on training and test sets, respectively. MAE = Mean Absolute Error; RMSE = Root Mean Square Error. CV R^2 represents mean $\pm$ standard deviation from 5-fold cross-validation. Optimal models: Biochar (Extra Trees, Test $R^2 = 0.887$), Bio-oil (XGBoost, Test $R^2 = 0.733$), Biogas (Ridge, Test $R^2 = 0.764$).

Trees exhibiting the narrowest error spread and demonstrating superior robustness across the full yield range.

Learning curves revealed data sufficiency and model behavior (Fig. 6). Extra Trees for biochar showed convergence of training and validation scores at ~ 100 samples (final $R^2 \approx 0.88$ -0.98), indicating adequate data for this product phase. XGBoost for bio-oil demonstrated stable performance with overlapping curves throughout, though absolute test R^2 (0.73) was lower than cross-validation R^2 (~ 0.85), suggesting distribution differences between training and test sets. Ridge regression for biogas exhibited declining validation performance with

increasing sample size (R^2 dropping from 0.95 to 0.75-0.80), characteristic of underfitting and indicating insufficient model complexity for capturing nonlinear biogas formation mechanisms. These results demonstrated that the optimized models achieved prediction errors within 5 wt% across all product phases, providing sufficient accuracy for practical process optimization and yield estimation in biomass-plastic co-pyrolysis systems [29–31].

3.6. Feature stability and heterogeneity analysis

SHAP analysis identified key predictive features across all three product phases (Fig. 7). For biochar, P_WC (plastic water content), Temperature, and B_FC (biomass fixed carbon) dominated, collectively explaining $>65\%$ of variance. Temperature showed bidirectional effects: moderate temperatures (400–600 °C) increased biochar yield through primary devolatilization, while high temperatures (>700 °C) decreased it via secondary gasification. Bio-oil formation was primarily controlled by Temperature (SHAP range: 10 to +15), exhibiting an inverted-U relationship with maximum yields at 500–650 °C, followed by H_C_Ratio_Biomass as the secondary driver. Biogas yield showed monotonic positive correlation with Temperature (SHAP range: 20 to +40), with P_O (plastic oxygen) and P_N (plastic nitrogen) as secondary factors influencing radical chemistry. The weak influence of Biomass_Plastic_Ratio on bio-oil suggested limited direct synergy for liquid product formation, with compositional effects mediated primarily through elemental ratios rather than mixing proportions [32–34].

Extended interpretability diagnostics verified the robustness of feature importance rankings across all three product phases. SHAP dependence plots (Fig. S8) showed consistent nonlinear patterns across multiple random seeds, confirming that identified relationships reflect genuine physicochemical effects rather than artifacts of model initialization. ICE plots (Fig. S9) revealed sample-specific heterogeneity: high-ash biomasses (>15 wt%) exhibited attenuated temperature sensitivity, likely due to catalytic mineral effects, while nitrogen-rich feedstocks (>3 wt% N) showed enhanced biogas yields at lower temperatures, indicating accelerated radical formation pathways. These subgroup-specific patterns demonstrate the value of sample-level interpretation for understanding composition-dependent reactivity in co-pyrolysis systems [35–37].

Dependence and interaction analyses revealed pronounced nonlinear responses of tri-phase yields to temperature, feed ratio, and composition (Fig. 8). Temperature exerted the dominant effect, with SHAP values increasing steeply between 400 and 650 °C - corresponding to the transition from primary devolatilization to secondary cracking - and reaching product-specific optima near 550 °C for biochar, 600 °C for bio-oil, and >700 °C for biogas. Biochar formation followed a sigmoidal temperature pattern and was maximized at moderate biomass-to-plastic

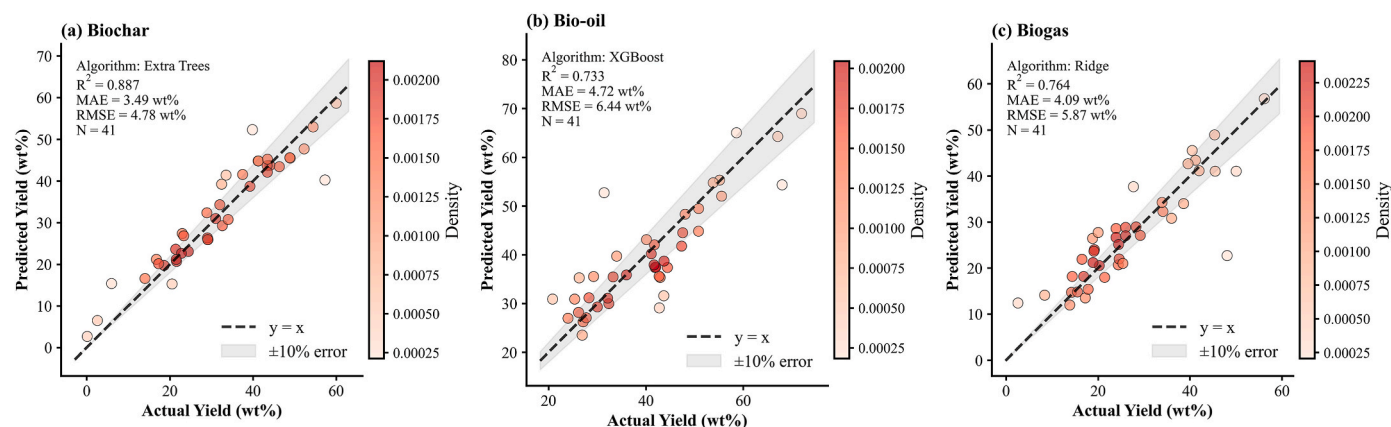


Fig. 5. Model prediction accuracy. Test set (N = 41) parity plots: (a) Biochar $R^2 = 0.887$; (b) Bio-oil $R^2 = 0.733$; (c) Biogas $R^2 = 0.764$. Dashed line = 1:1. Gray band = $\pm 10\%$ error.

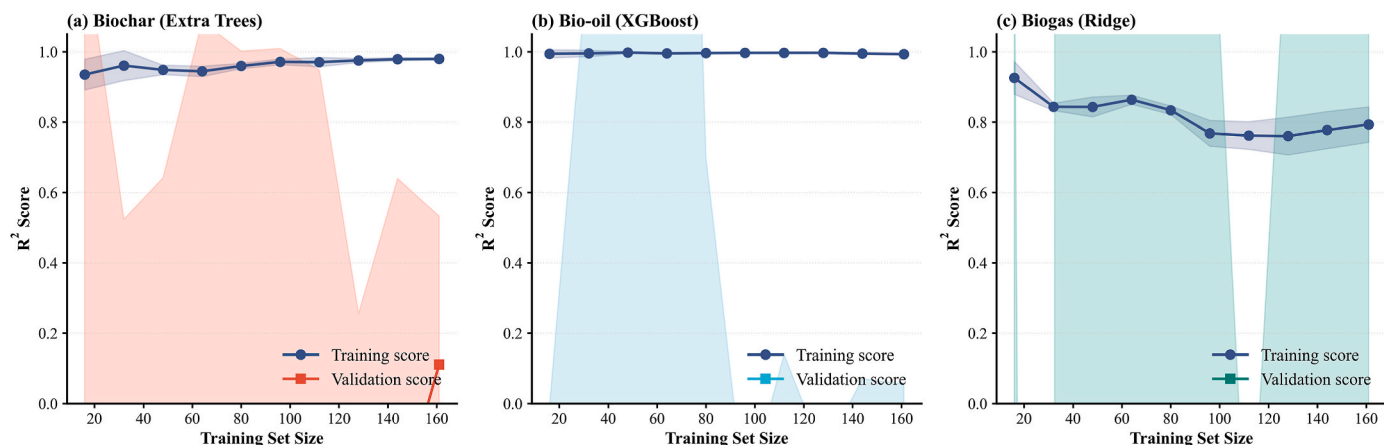


Fig. 6. Learning curves. Training and validation R^2 for optimal models: (a) biochar, (b) bio-oil, (c) biogas.

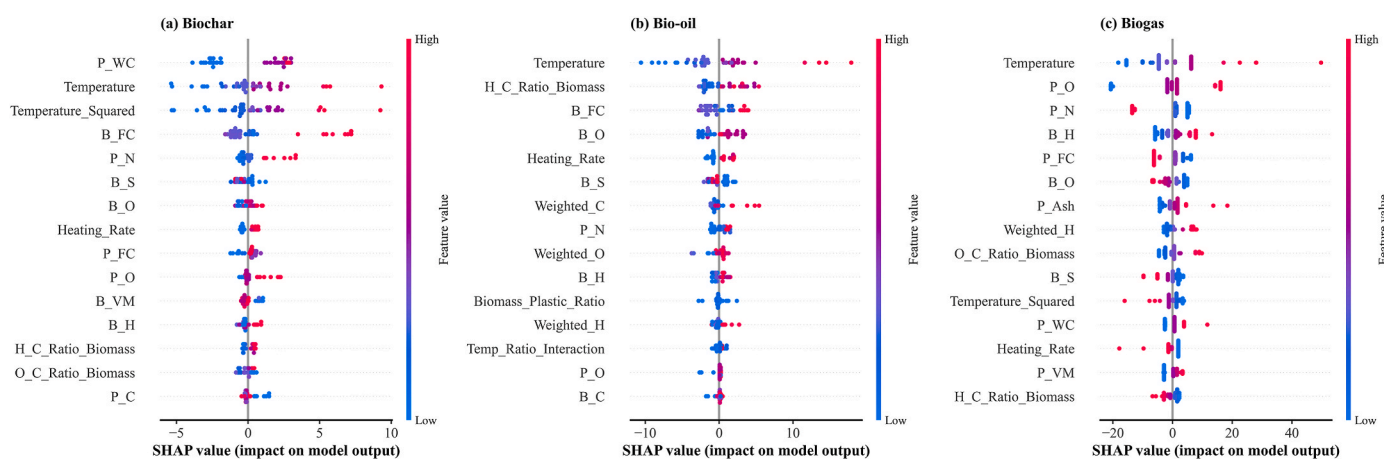


Fig. 7. SHAP feature importance. Top predictive features for (a) biochar, (b) bio-oil, and (c) biogas ranked by mean absolute SHAP values.

ratios (0.3–0.7), confirming the presence of co-pyrolysis synergy. In contrast, extreme blend ratios or sulfur-rich feedstocks suppressed char yield through radical-mediated decomposition pathways. Bio-oil displayed an inverted-U-shaped temperature response, indicating a well-defined window for liquid production; plastic-rich blends favored oil yield due to their high H/C ratios and aliphatic structure, though ash minerals introduced catalytic variability that broadened the optimal temperature range. Biogas formation increased exponentially with temperature and with higher plastic content and biomass carbon content, reflecting kinetically driven cracking reactions and methane generation from carbonaceous precursors [38–41].

3.7. Interaction mapping of synergistic windows

Temperature-ratio interaction heatmaps (Fig. S10) quantified synergistic regions where co-pyrolysis yields exceeded weighted averages of individual feedstock conversion. Three distinct operational windows emerged: (1) biochar maximization at 500–550 °C with 30–70% biomass, achieving 5–15% yield enhancement over predicted values; (2) bio-oil optimization at 550–650 °C with 40–60% biomass, showing 8–12% synergistic gains; and (3) biogas generation at >700 °C with <40% biomass, exhibiting exponential yield increases. These interaction patterns provide quantitative evidence of coupled thermal-compositional controls and define process conditions for targeted product selectivity [42–45].

Partial-dependence analysis quantified average marginal sensitivities of key process variables on product yields (Fig. 9). Temperature

exhibited the steepest slope for bio-oil ($\partial Y/\partial T \approx 0.12 \text{ wt}\% \cdot ^\circ\text{C}^{-1}$ below 600 °C) and a negative gradient for biochar ($-0.09 \text{ wt}\% \cdot ^\circ\text{C}^{-1}$), while biomass/plastic ratio had a stronger effect on liquids ($\partial Y/\partial R \approx 0.18$) than on gases (0.07). Within the optimal operational window (500–650 °C; 40–60% biomass), these gradients were approximately quasi-linear, consistent with the nonlinear thresholds identified by SHAP and ICE analyses. Averaged PDPs showed nearly flat temperature responses for biochar and bio-oil at the population level, indicating that large sample-specific effects cancel out in aggregate, whereas feedstock-specific variation prevents identification of a universal optimum. Biogas yield increased monotonically with heating rate, confirming kinetic rather than equilibrium control of gas-phase reactions.

3.8. Integrated interpretability framework

Collectively, the comparison among SHAP (local feature attribution), PDP (population-averaged effects), and ICE (sample-specific responses) highlighted that co-pyrolysis behavior is dominated by interaction effects rather than single-factor influences. The complementary evidence from main-text figures and supporting information (Figs. S3–S10, Tables S2–S3) strengthens the reliability of the established machine-learning framework through multi-level validation of robustness, interaction dynamics, and mechanistic consistency. Ensemble models such as Extra Trees and XGBoost effectively capture these coupled dependencies, achieving $R^2 \approx 0.89$ while maintaining mechanistic interpretability. The combined SHAP-PDP-ICE framework delineated clear operational windows (500–550 °C for biochar, 550–650 °C for bio-oil,

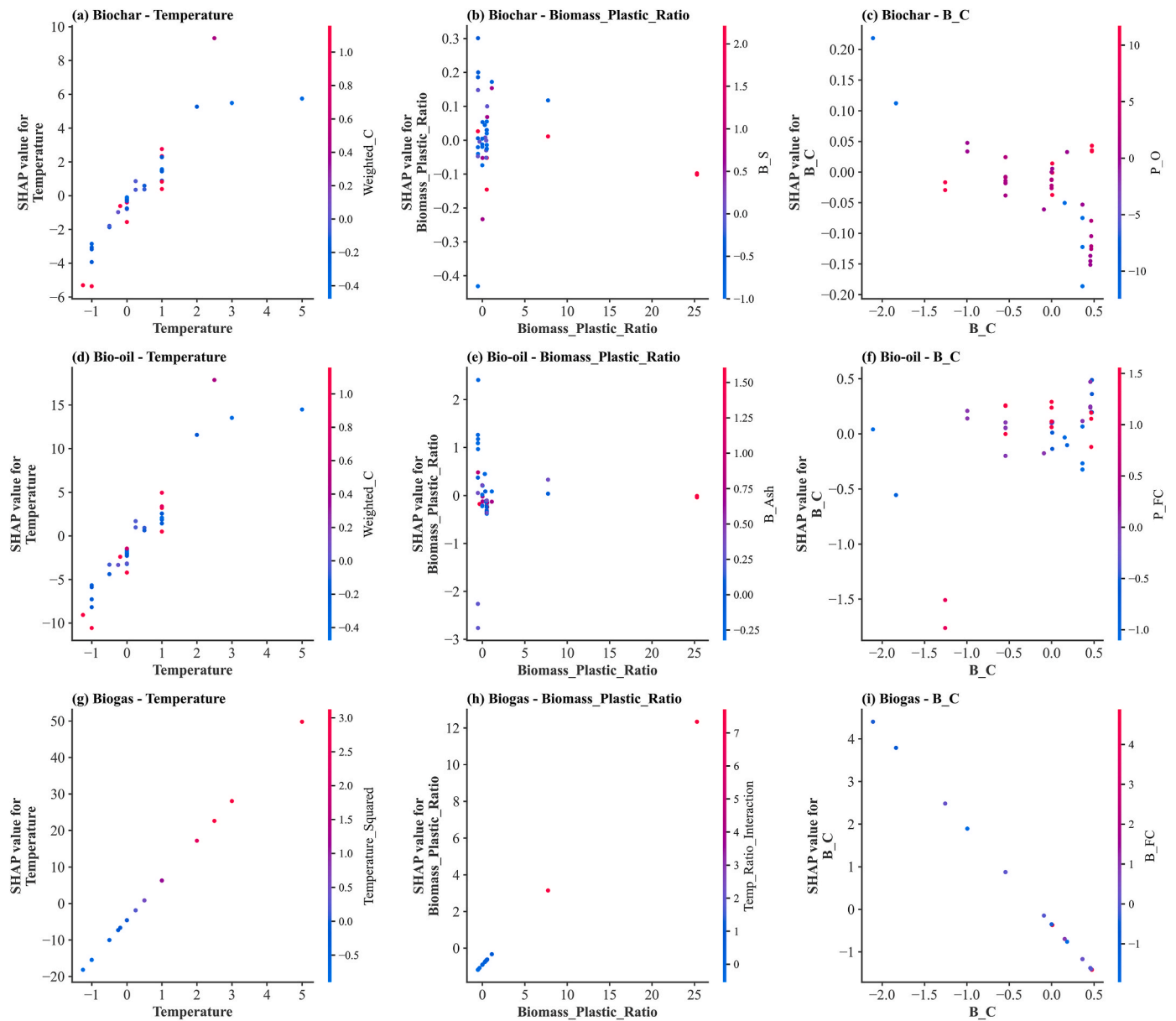


Fig. 8. SHAP dependence plots. Individual feature effects for top predictors of (a-c) biochar, (d-f) bio-oil, and (g-i) biogas.

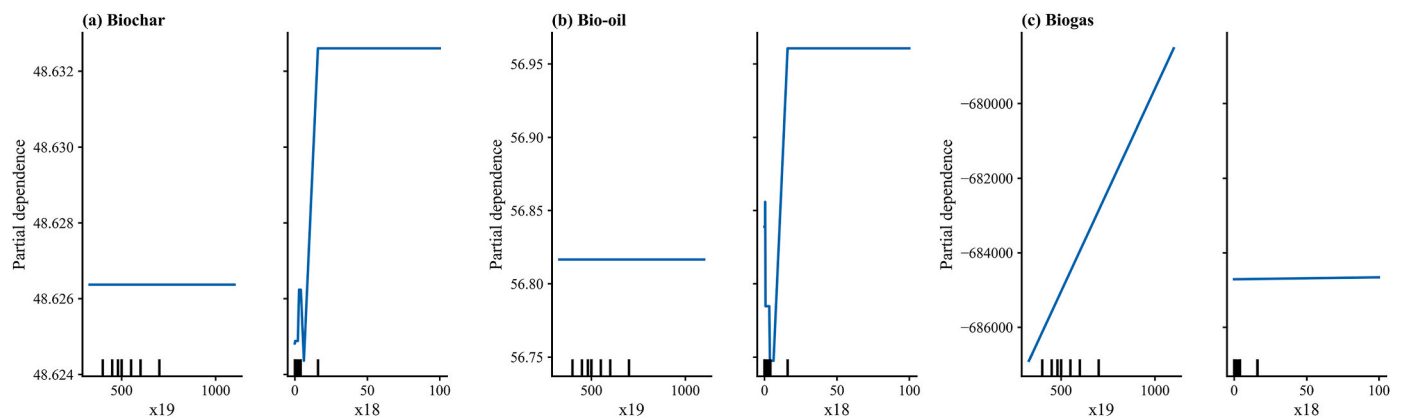


Fig. 9. Partial dependence plots of temperature and heating rate effects on biochar, bio-oil, and biogas yields.

and above 700 °C for biogas) and provided a quantitative basis for data-driven process optimization in waste valorization applications [46–49].

Despite encompassing five reactor configurations (Fig. S11), the ensemble algorithms demonstrated robust cross-validation performance ($R^2 > 0.75$ for all products), suggesting that process parameters (temperature, heating rate) and feedstock properties effectively capture reactor-independent thermochemical relationships. While reactor configuration could influence product yields by 5–15% through variations in heat transfer and residence time, the strong model performance demonstrated that temperature and feedstock composition remained the dominant variables within the studied operational windows. This study focused on yield prediction as the primary economic driver, restricting the scope to thermal (non-catalytic) pyrolysis to maintain dataset homogeneity. Product quality data and catalytic process data were excluded, as these parameters are inconsistently reported across literature sources, rendering comprehensive modeling infeasible with the current dataset. Future work could explore reactor-specific models, incorporate reactor type as a categorical feature, develop integrated yield-quality models, and extend to catalytic co-pyrolysis once standardized datasets become available.

4. Conclusion

This study establishes an interpretable machine-learning framework for predicting tri-phase product distributions in biomass-plastic co-pyrolysis. Product-specific algorithm selection - Extra Trees for biochar, XGBoost for bio-oil, and Ridge regression for biogas - achieved high accuracy (R^2 up to 0.89) across 202 experimental datasets spanning diverse feedstocks and conditions. SHAP and partial dependence analyses identified temperature and biomass-to-plastic ratio as the dominant control variables, with hydrogen-to-carbon ratio and ash content as secondary modulators. Interaction mapping defined operational optima at 500–550 °C for biochar, 550–650 °C for bio-oil, and above 700 °C for biogas, linking the observed synergistic effects to polymer hydrogen donation, biomass radical quenching, and secondary cracking mechanisms. Beyond predictive accuracy, this framework demonstrated how explainable AI tools can bridge empirical data and thermochemical understanding, transforming conventional black-box models into interpretable decision-support systems. The approach is readily extensible to catalytic co-pyrolysis, multi-feedstock optimization, and real-time process control. Future work should integrate kinetic modeling with data-driven predictions and validate framework transferability across reactor configurations and scales, advancing intelligent waste valorization toward circular economy targets.

CRediT authorship contribution statement

Jiawei Tan: Writing – original draft, Software, Resources, Investigation. **Qicheng Lai:** Methodology, Investigation, Data curation, Conceptualization. **Yuanbo Song:** Writing – review & editing, Supervision, Resources. **Xiang Qin:** Validation, Resources, Investigation. **Junhao Hu:** Visualization, Investigation, Data curation. **Pan Li:** Writing – original draft, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare no competing financial interest.

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Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2026.140729>.

Data availability

Data will be made available on request.

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